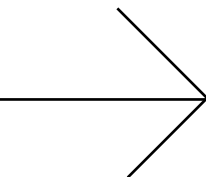
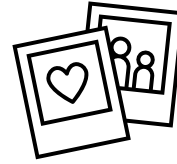


# Disentangling the role of semantic prior and sensory details in autobiographical memory retrieval

Uyen Doan & Prof. Andrew Perfors  
Complex Human Data Hub  
The University of Melbourne

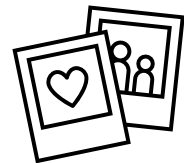


# Autobiographical memory retrieval



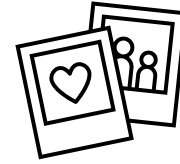
| Example: a memory in relation to the word <u>calm</u>                                                                                                                                                                 | Memory type |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| after one chaotic day of exams, I went to the lake near campus at sunset, the water was completely still and no one else was there just sitting on the dock listening to the ripples made my whole body finally relax |             |
|                                                                                                                                                                                                                       |             |
|                                                                                                                                                                                                                       |             |
|                                                                                                                                                                                                                       |             |

# Autobiographical memory retrieval



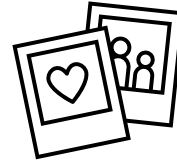
| Example: a memory in relation to the word <u>calm</u>                                                                                                                                                                 | Memory type |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| after one chaotic day of exams, I went to the lake near campus at sunset, the water was completely still and no one else was there just sitting on the dock listening to the ripples made my whole body finally relax | Specific    |
|                                                                                                                                                                                                                       |             |
|                                                                                                                                                                                                                       |             |
|                                                                                                                                                                                                                       |             |

# Autobiographical memory retrieval



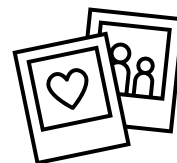
| Example: a memory in relation to the word <u>calm</u>                                                                                                                                                                 | Memory type |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| after one chaotic day of exams, I went to the lake near campus at sunset, the water was completely still and no one else was there just sitting on the dock listening to the ripples made my whole body finally relax | Specific    |
| I grew up in Corpus Christi, so we had the beach close by and sitting on the beach, listening to the waves is one of the calm experiences that you can encounter or experience in life, relaxing very peaceful.       | Extended    |
|                                                                                                                                                                                                                       |             |
|                                                                                                                                                                                                                       |             |

# Autobiographical memory retrieval



| Example: a memory in relation to the word <u>calm</u>                                                                                                                                                                 | Memory type        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| after one chaotic day of exams, I went to the lake near campus at sunset, the water was completely still and no one else was there just sitting on the dock listening to the ripples made my whole body finally relax | Specific           |
| I grew up in Corpus Christi, so we had the beach close by and sitting on the beach, listening to the waves is one of the calm experiences that you can encounter or experience in life, relaxing very peaceful.       | Extended           |
| Curled up in a chair with my cat Sammy on Sunday mornings, petting him and his sweet face looking at me with love.                                                                                                    | Categoric          |
| Moms voice                                                                                                                                                                                                            | Semantic associate |

# Autobiographical memory retrieval

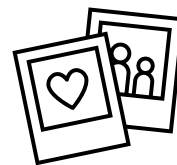


| Example: a memory in relation to the word <u>calm</u>                                                                                                                                                                 | Memory type        | specific |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|----------|
| after one chaotic day of exams, I went to the lake near campus at sunset, the water was completely still and no one else was there just sitting on the dock listening to the ripples made my whole body finally relax | Specific           |          |
| I grew up in Corpus Christi, so we had the beach close by and sitting on the beach, listening to the waves is one of the calm experiences that you can encounter or experience in life, relaxing very peaceful.       | Extended           |          |
| Curled up in a chair with my cat Sammy on Sunday mornings, petting him and his sweet face looking at me with love.                                                                                                    | Categoric          | general  |
| Moms voice                                                                                                                                                                                                            | Semantic associate |          |

Some people tend to give general summaries of events, even when asked for unique specific events.

Consistent over time, across different cues (e.g., Gutenbrunner et al., 2018; King et al., 2023; Ridout et al., 2016, Sumner et al., 2013)

# Autobiographical memory retrieval



| Example: a memory in relation to the word <u>calm</u>                                                                                                                                                                 | Memory type        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| after one chaotic day of exams, I went to the lake near campus at sunset, the water was completely still and no one else was there just sitting on the dock listening to the ripples made my whole body finally relax | Specific           |
| I grew up in Corpus Christi, so we had the beach close by and sitting on the beach, listening to the waves is one of the calm experiences that you can encounter or experience in life, relaxing very peaceful.       | Extended           |
| Curled up in a chair with my cat Sammy on Sunday mornings, petting him and his sweet face looking at me with love.                                                                                                    | Categoric          |
| Moms voice                                                                                                                                                                                                            | Semantic associate |

episodic

semantic

Some people tend to give general summaries of events, even when asked for unique specific events.

Consistent over time, across different cues (e.g., Gutenbrunner et al., 2018; King et al., 2023; Ridout et al., 2016, Sumner et al., 2013)

# Memory specificity is related to many aspects of psychological functioning

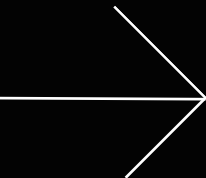
In daily functioning, more specific memory is associated with:

- Effectiveness in problem solving (Dristchel et al., 1996; Jing et al., 2016)
- Planning and imagining future events (Benoit & Schacter, 2015; Jing et al., 2017)
- Social sharing and bonding (Beike et al., 2016; Barry et al., 2019)

In psychopathology, less specific memory is considered:

- Transdiagnostic cognitive marker of psychiatric disorders (for reviews see Barry et al., 2021)
- Risk factor for future depressive symptoms and onset (for reviews see Halford et al., 2021; Doan et al., 2025)

But why? And what is it  
measuring, precisely?



# How are semantic and episodic memories related?

One recently proposed framework: adaptively compressed episodic and semantic memories help solve online learning problem (Nagy, Orban, & Wu, 2025)

- **Episodic** memories are characterised as **sensory traces** of unique event
- **Semantic** memories are **gist-based model** of the environment, a probabilistic generative model over past experiences

**Semantic** memory provides a model for new events to be either efficiently compressed if fully predicted by the model while **surprising** information are stored relatively uncompressed and **update** existing model.

This framework relates memory processes to both solving the problem of learning the model first, and, of perception when the model is applied to new information.

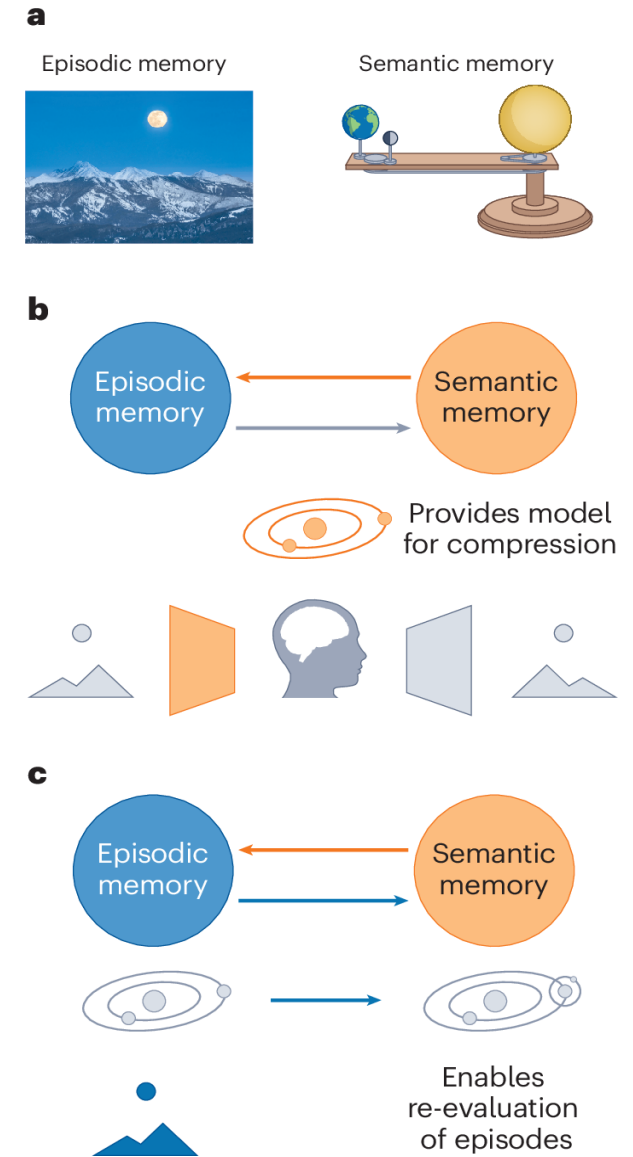
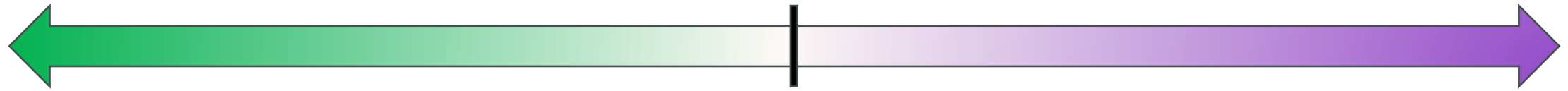


Figure from Nagy et al., 2025

# We still need an explanation for individual differences

Individuals may differ in how they make the trade-off between:



Semantic gist

Sensory details

General memory

Specific memory

Prior knowledge

Sensory evidence

Higher error threshold leads to less sensory information encoded/retrieved, and the generative model shows more gist-based distortion but is more efficient (Spens & Burgess, 2023).

Bringing  
these  
ideas  
together...

**What if autobiographical memory specificity partly captures this broader trade-off in the domain of memory?**

Maybe people who rely more on prior knowledge retrieve more summaries of events and encode/retrieve less sensory details when encountering surprising events.

**Is this reliance on prior domain-general?**

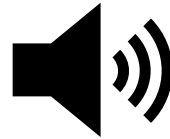
Previous work on autobiographical memory hasn't explored this possibility as it is mainly focused on clinical implications and factors like attention, motivation, and emotion

# How can we test this?

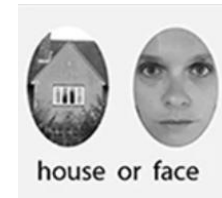
By comparing people's memory specificity as measured by the AMT to their performance on very different tasks which each captures how that person trades off between prior knowledge and sensory input



Autobiographical  
memory test



Sentence  
perception task

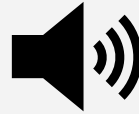


Probabilistic  
learning task

# DESIGN

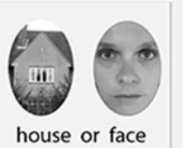
## Session One

- AMT – minimal instruction (12 cues)
- Sentence perception task



## Session Two

- AMT – original instruction (12 cues)
- Probabilistic learning task

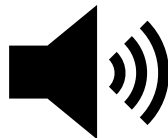


# SAMPLE

- $n = 120$ ,  $M_{\text{age}} = 44$
- Native English speakers
- US-based

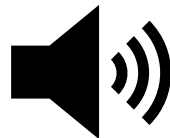
- $n = 98$
- 2 days after Session One

# SENTENCE PERCEPTION TASK



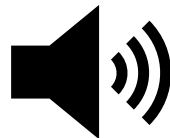
|  | Semantic Prior<br><i>Entropy</i>                          |  |  |  |  |  |  |  |
|--|-----------------------------------------------------------|--|--|--|--|--|--|--|
|  |                                                           |  |  |  |  |  |  |  |
|  | After its injury, the small bird could no longer<br>(0.1) |  |  |  |  |  |  |  |
|  |                                                           |  |  |  |  |  |  |  |
|  |                                                           |  |  |  |  |  |  |  |
|  |                                                           |  |  |  |  |  |  |  |
|  |                                                           |  |  |  |  |  |  |  |

# SENTENCE PERCEPTION TASK



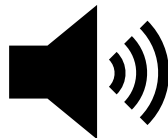
|  | Semantic Prior                                            | Sensory Evidence |  |  |  |  |  |
|--|-----------------------------------------------------------|------------------|--|--|--|--|--|
|  | <i>Entropy</i>                                            | <i>Cloze</i>     |  |  |  |  |  |
|  | After its injury, the small bird could no longer<br>(0.1) | fly<br>(0.99)    |  |  |  |  |  |
|  |                                                           |                  |  |  |  |  |  |
|  |                                                           |                  |  |  |  |  |  |
|  |                                                           |                  |  |  |  |  |  |
|  |                                                           |                  |  |  |  |  |  |

# SENTENCE PERCEPTION TASK



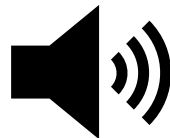
|  | Semantic Prior                                            | Sensory Evidence |                            |   |   |   |  |  |
|--|-----------------------------------------------------------|------------------|----------------------------|---|---|---|--|--|
|  | <i>Entropy</i>                                            | <i>Cloze</i>     | <i>Sensory degradation</i> |   |   |   |  |  |
|  | After its injury, the small bird could no longer<br>(0.1) | fly<br>(0.99)    | 12                         | 6 | 3 | 1 |  |  |
|  |                                                           |                  |                            |   |   |   |  |  |
|  |                                                           |                  |                            |   |   |   |  |  |
|  |                                                           |                  |                            |   |   |   |  |  |
|  |                                                           |                  |                            |   |   |   |  |  |

# SENTENCE PERCEPTION TASK



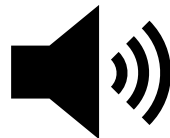
| Condition                   | Semantic Prior<br><i>Entropy</i>                          | Sensory Evidence |                            |   |   |   | Clarity<br>Rating | Posterior  |
|-----------------------------|-----------------------------------------------------------|------------------|----------------------------|---|---|---|-------------------|------------|
|                             |                                                           | <i>Cloze</i>     | <i>Sensory degradation</i> |   |   |   |                   |            |
| Low Entropy - E<br>(n = 48) | After its injury, the small bird could no longer<br>(0.1) | fly<br>(0.99)    | 12                         | 6 | 3 | 1 | 1-100             | [text box] |
|                             |                                                           |                  |                            |   |   |   |                   |            |
|                             |                                                           |                  |                            |   |   |   |                   |            |
|                             |                                                           |                  |                            |   |   |   |                   |            |
|                             |                                                           |                  |                            |   |   |   |                   |            |

# SENTENCE PERCEPTION TASK



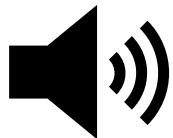
| Condition                    | Semantic Prior<br><i>Entropy</i>                          | Sensory Evidence |                            |   |   |   | Clarity<br>Rating | Posterior  |
|------------------------------|-----------------------------------------------------------|------------------|----------------------------|---|---|---|-------------------|------------|
|                              |                                                           | <i>Cloze</i>     | <i>Sensory degradation</i> |   |   |   |                   |            |
| Low Entropy - E<br>(n = 48)  | After its injury, the small bird could no longer<br>(0.1) | fly<br>(0.99)    | 12                         | 6 | 3 | 1 | 1-100             | [text box] |
| Low Entropy - UE<br>(n = 48) | After its injury, the small bird could no longer<br>(0.1) | sing<br>(0.0)    |                            |   |   |   |                   |            |
|                              |                                                           |                  |                            |   |   |   |                   |            |
|                              |                                                           |                  |                            |   |   |   |                   |            |
|                              |                                                           |                  |                            |   |   |   |                   |            |

# SENTENCE PERCEPTION TASK



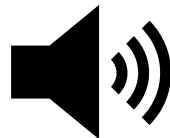
| Condition                    | Semantic Prior<br><i>Entropy</i>                          | Sensory Evidence |                            |    |   |   | Clarity<br>Rating | Posterior  |
|------------------------------|-----------------------------------------------------------|------------------|----------------------------|----|---|---|-------------------|------------|
|                              |                                                           | <i>Cloze</i>     | <i>Sensory degradation</i> |    |   |   |                   |            |
| Low Entropy - E<br>(n = 48)  | After its injury, the small bird could no longer<br>(0.1) | fly<br>(0.99)    | 12                         | 6  | 3 | 1 | 1-100             | [text box] |
| Low Entropy - UE<br>(n = 48) | After its injury, the small bird could no longer<br>(0.1) | sing<br>(0.0)    | 1                          | 12 | 6 | 3 | 1-100             | [text box] |
|                              |                                                           |                  |                            |    |   |   |                   |            |
|                              |                                                           |                  |                            |    |   |   |                   |            |
|                              |                                                           |                  |                            |    |   |   |                   |            |

# SENTENCE PERCEPTION TASK



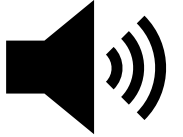
| Condition                      | Semantic Prior<br><i>Entropy</i>                          | Sensory Evidence |                            |    |    |   | Clarity<br>Rating | Posterior  |
|--------------------------------|-----------------------------------------------------------|------------------|----------------------------|----|----|---|-------------------|------------|
|                                |                                                           | <i>Cloze</i>     | <i>Sensory degradation</i> |    |    |   |                   |            |
| Low Entropy - E<br>(n = 48)    | After its injury, the small bird could no longer<br>(0.1) | fly<br>(0.99)    | 12                         | 6  | 3  | 1 | 1-100             | [text box] |
| Low Entropy - UE<br>(n = 48)   | After its injury, the small bird could no longer<br>(0.1) | sing<br>(0.0)    | 1                          | 12 | 6  | 3 | 1-100             | [text box] |
| Medium Entropy - E<br>(n = 48) | The driver said the best flavour of pie is<br>(1.8)       | apple<br>(0.5)   | 3                          | 1  | 12 | 6 | 1-100             | [text box] |
|                                |                                                           |                  |                            |    |    |   |                   |            |
|                                |                                                           |                  |                            |    |    |   |                   |            |

# SENTENCE PERCEPTION TASK



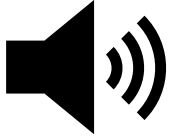
| Condition                       | Semantic Prior<br><i>Entropy</i>                          | Sensory Evidence  |                            |    |    |    | Clarity<br>Rating | Posterior  |
|---------------------------------|-----------------------------------------------------------|-------------------|----------------------------|----|----|----|-------------------|------------|
|                                 |                                                           | <i>Cloze</i>      | <i>Sensory degradation</i> |    |    |    |                   |            |
| Low Entropy - E<br>(n = 48)     | After its injury, the small bird could no longer<br>(0.1) | fly<br>(0.99)     | 12                         | 6  | 3  | 1  | 1-100             | [text box] |
| Low Entropy - UE<br>(n = 48)    | After its injury, the small bird could no longer<br>(0.1) | sing<br>(0.0)     | 1                          | 12 | 6  | 3  | 1-100             | [text box] |
| Medium Entropy - E<br>(n = 48)  | The driver said the best flavour of pie is<br>(1.8)       | apple<br>(0.5)    | 3                          | 1  | 12 | 6  | 1-100             | [text box] |
| Medium Entropy - UE<br>(n = 48) | The driver said the best flavour of pie is<br>(1.8)       | pumpkin<br>(0.02) | 6                          | 3  | 1  | 12 | 1-100             | [text box] |
|                                 |                                                           |                   |                            |    |    |    |                   |            |

# SENTENCE PERCEPTION TASK



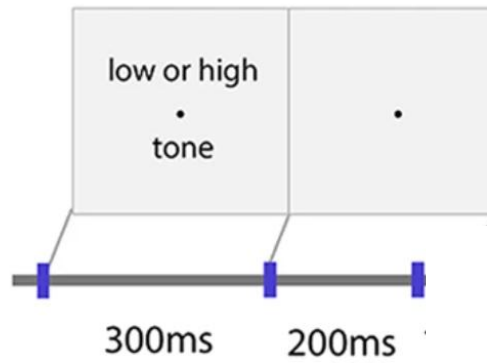
| Condition                       | Semantic Prior                                               | Sensory Evidence  |                            |    |    |    | Clarity Rating | Posterior  |
|---------------------------------|--------------------------------------------------------------|-------------------|----------------------------|----|----|----|----------------|------------|
|                                 | <i>Entropy</i>                                               | <i>Cloze</i>      | <i>Sensory degradation</i> |    |    |    |                |            |
| Low Entropy - E<br>(n = 48)     | After its injury, the small bird could no longer<br>(0.1)    | fly<br>(0.99)     | 12                         | 6  | 3  | 1  | 1-100          | [text box] |
| Low Entropy - UE<br>(n = 48)    | After its injury, the small bird could no longer<br>(0.1)    | sing<br>(0.0)     | 1                          | 12 | 6  | 3  | 1-100          | [text box] |
| Medium Entropy - E<br>(n = 48)  | The driver said the best flavour of pie is<br>(1.8)          | apple<br>(0.5)    | 3                          | 1  | 12 | 6  | 1-100          | [text box] |
| Medium Entropy - UE<br>(n = 48) | The driver said the best flavour of pie is<br>(1.8)          | pumpkin<br>(0.02) | 6                          | 3  | 1  | 12 | 1-100          | [text box] |
| High Entropy - E<br>(n = 48)    | The doctor was worried because the patient hurt his<br>(3.0) | leg<br>(0.2)      | 12                         | 6  | 3  | 1  | 1-100          | [text box] |

# SENTENCE PERCEPTION TASK

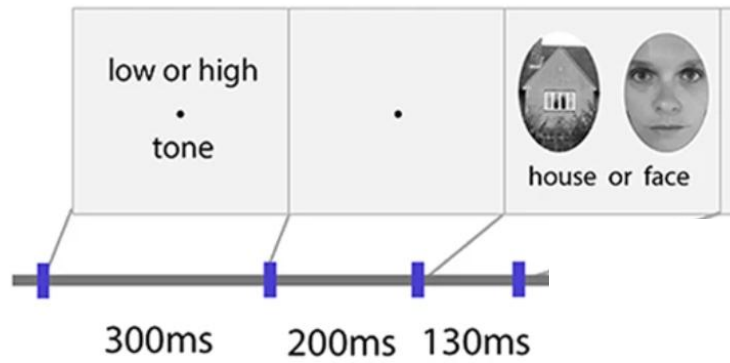


| Condition                       | Semantic Prior<br><i>Entropy</i>                             | Sensory Evidence  |                            |    |    |    | Clarity<br>Rating | Posterior  |
|---------------------------------|--------------------------------------------------------------|-------------------|----------------------------|----|----|----|-------------------|------------|
|                                 |                                                              | <i>Cloze</i>      | <i>Sensory degradation</i> |    |    |    |                   |            |
| Low Entropy - E<br>(n = 48)     | After its injury, the small bird could no longer<br>(0.1)    | fly<br>(0.99)     | 12                         | 6  | 3  | 1  | 1-100             | [text box] |
| Low Entropy - UE<br>(n = 48)    | After its injury, the small bird could no longer<br>(0.1)    | sing<br>(0.0)     | 1                          | 12 | 6  | 3  | 1-100             | [text box] |
| Medium Entropy - E<br>(n = 48)  | The driver said the best flavour of pie is<br>(1.8)          | apple<br>(0.5)    | 3                          | 1  | 12 | 6  | 1-100             | [text box] |
| Medium Entropy - UE<br>(n = 48) | The driver said the best flavour of pie is<br>(1.8)          | pumpkin<br>(0.02) | 6                          | 3  | 1  | 12 | 1-100             | [text box] |
| High Entropy - E<br>(n = 48)    | The doctor was worried because the patient hurt his<br>(3.0) | leg<br>(0.2)      | 12                         | 6  | 3  | 1  | 1-100             | [text box] |

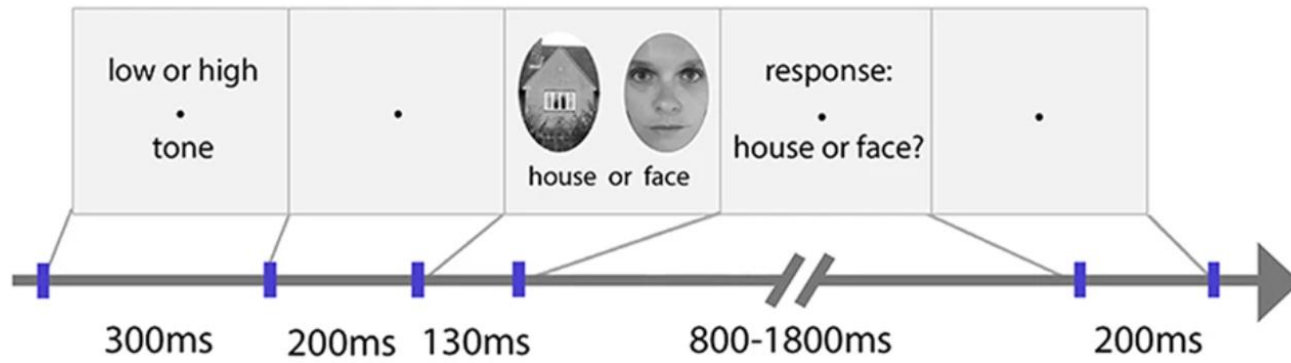
# PROBABILISTIC LEARNING TASK



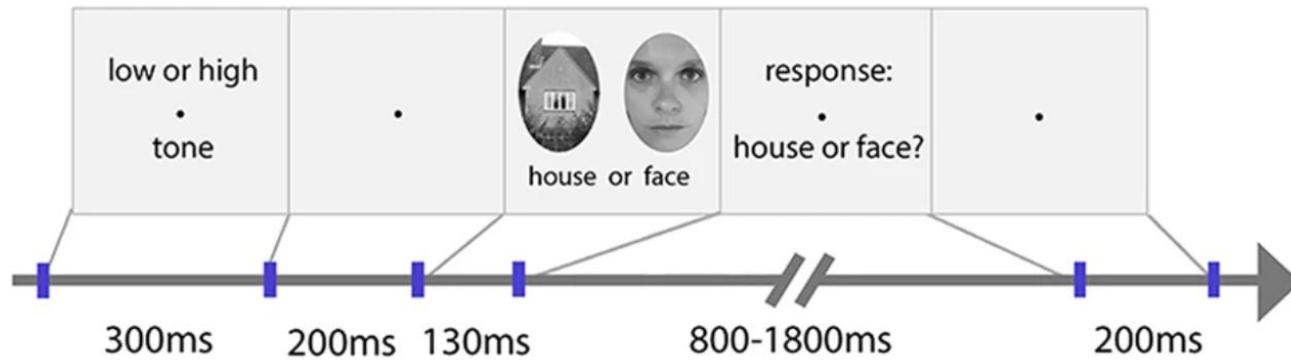
# PROBABILISTIC LEARNING TASK



# PROBABILISTIC LEARNING TASK



# PROBABILISTIC LEARNING TASK

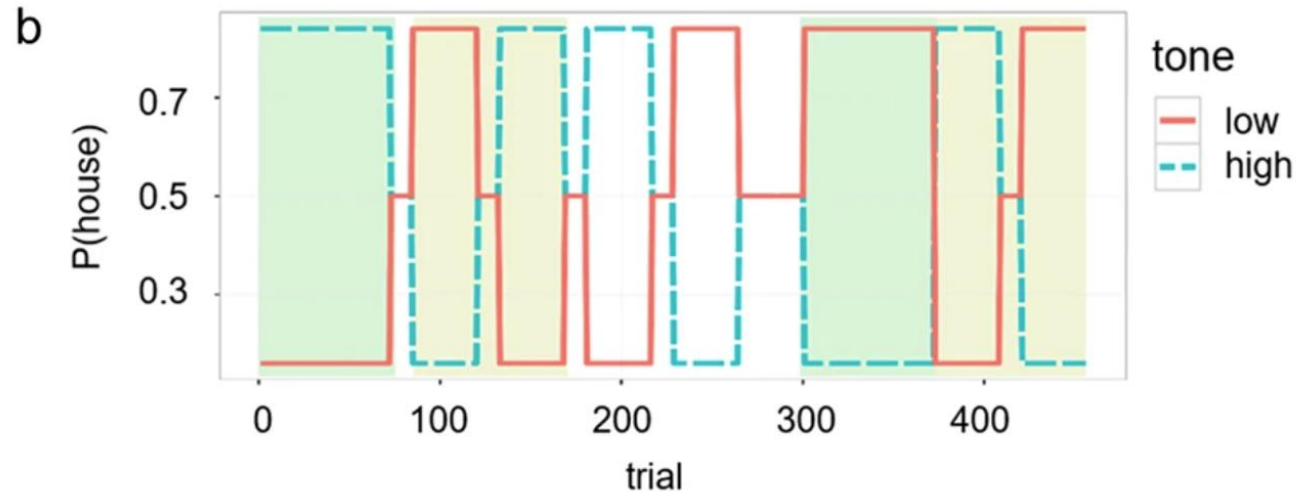


Prior: learn in task

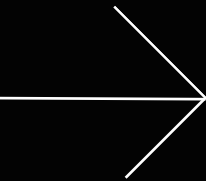
Sensory evidence: visual noise in image

Surprise: expectedness of tone – image pair

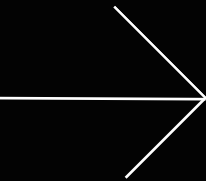
Outcome: trial response, RT



What did we find?

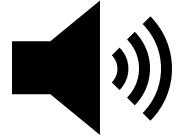


First, some sanity checks



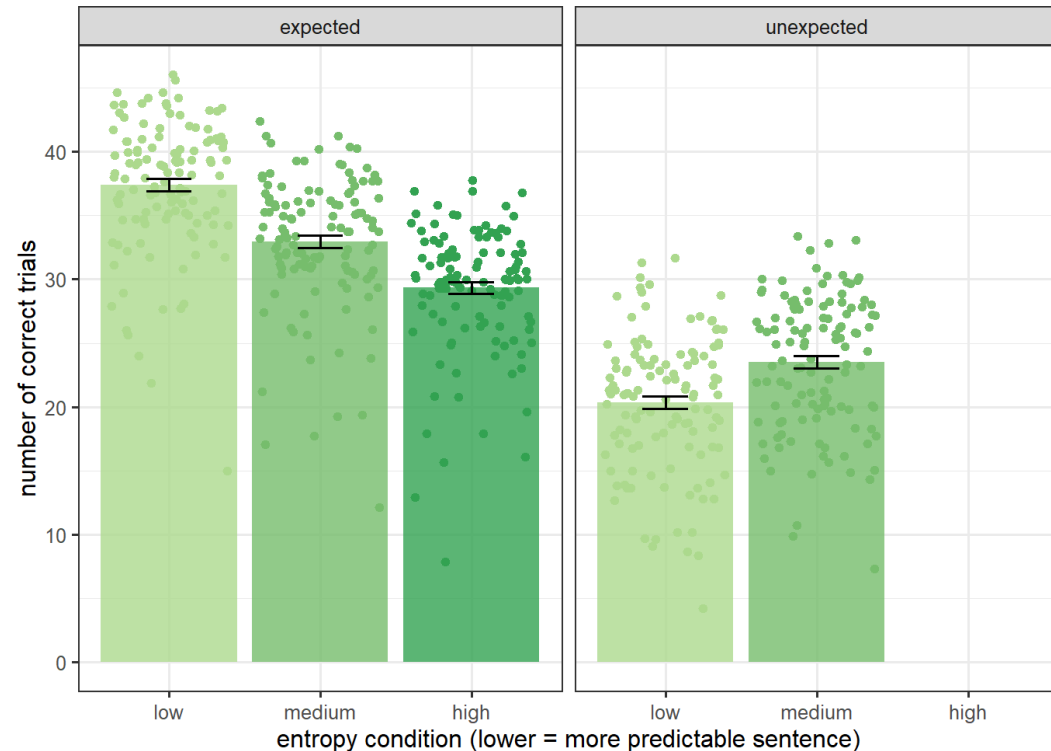
# Manipulation checks

People are more correct when the sentences are more predictable



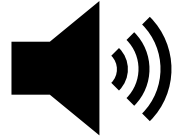
## Proportion of correct trials by predictability

Expected/ unexpected: last word matches sentence or surprising



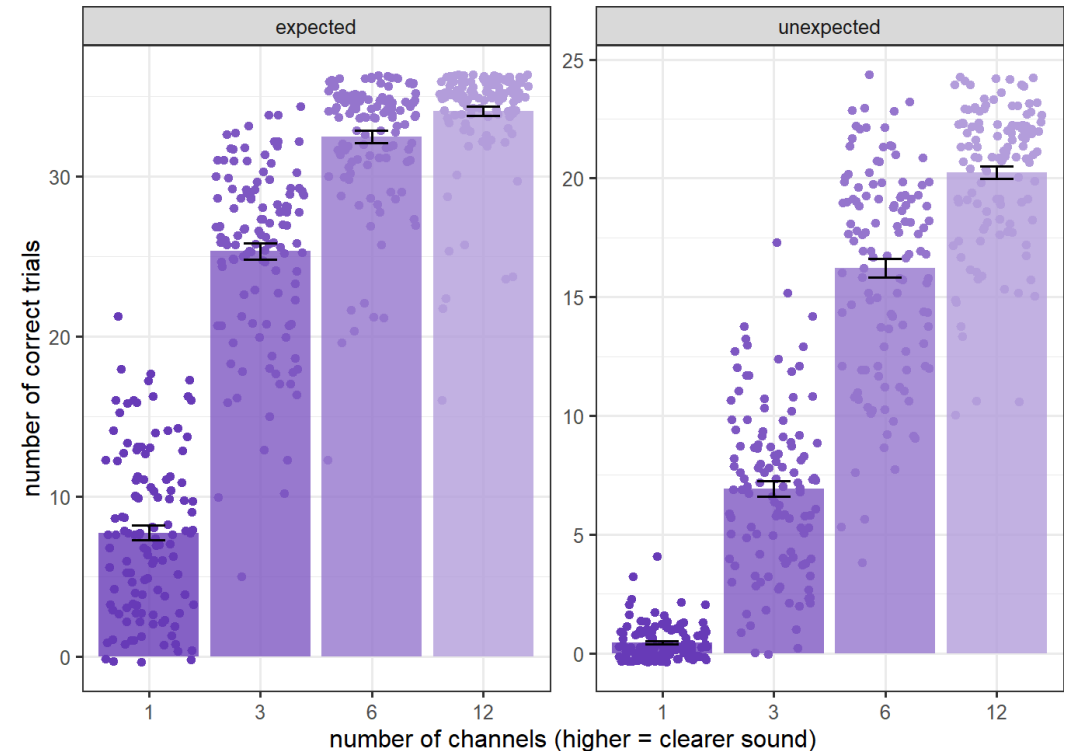
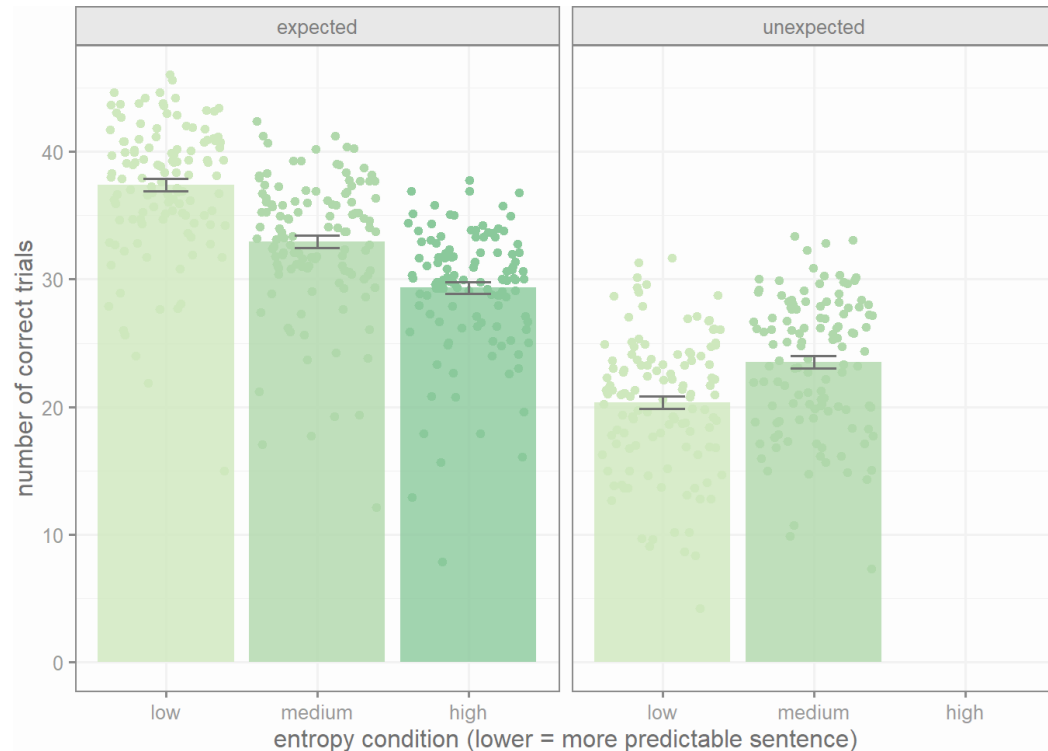
# Manipulation checks

People are more correct when the sound are less degraded



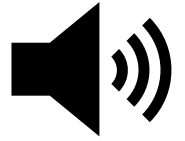
Proportion of correct trials by predictability and sound degradation

Expected/ unexpected: last word matches sentence or surprising



# Measuring reliance on prior

for each participant we fit a parameter reflecting this reliance



Linear model for each participant

$$\log(v_i) = \beta_1 + \beta_2 \times channels_i + \beta_3 \times entropy_i + \beta_4 \times cloze_i + \beta_5 \times clarity_i$$

Belief updating at each trial  $i$

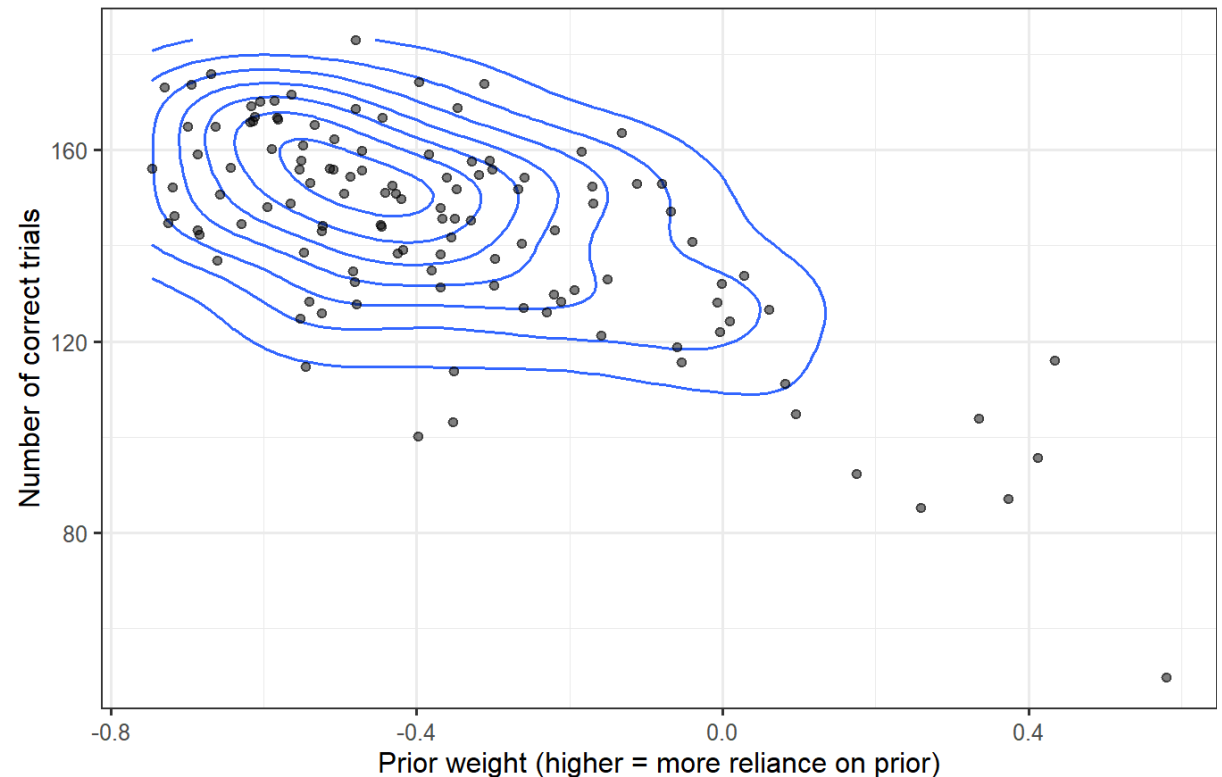
$$P_{m|n} = P_m + \frac{1}{v+1} (\delta_{m,n} - P_m)$$

Diagram illustrating the belief updating equation:

- $P_{m|n}$  is labeled as the **posterior**.
- $P_m$  is labeled as the **prior of response  $m$** .
- The term  $\frac{1}{v+1}$  is labeled as the **prior weight**.
- The term  $(\delta_{m,n} - P_m)$  is labeled as the **prediction error**.

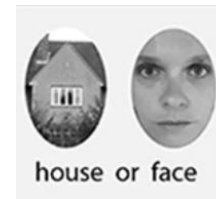
Density plot of prior weight and overall accuracy

Prior weight negatively correlated with accurate perception

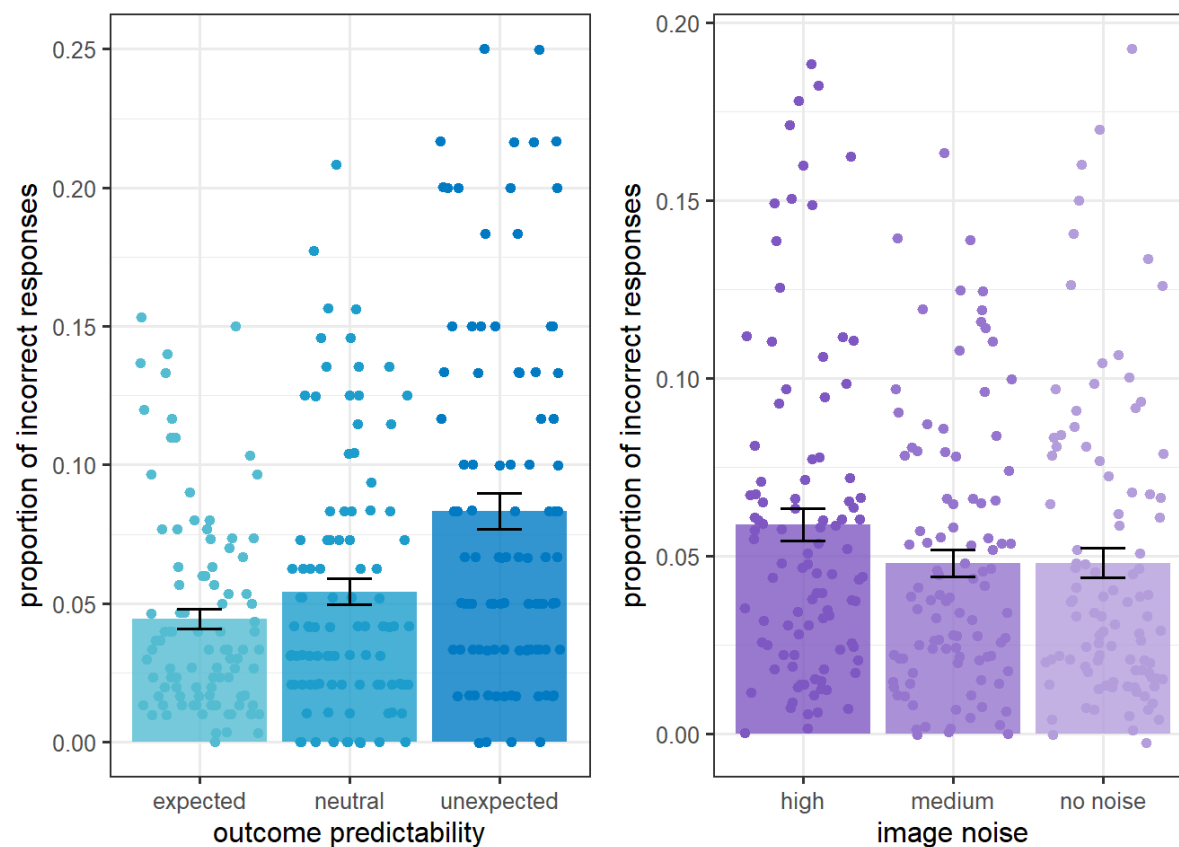


# Manipulation checks

## Learning task



Proportion of incorrect trials by predictability and image noise  
 $p(\text{outcome}|\text{cue}) = 0.83$  (expected),  $0.5$  (neutral),  $0.17$  (unexpected)



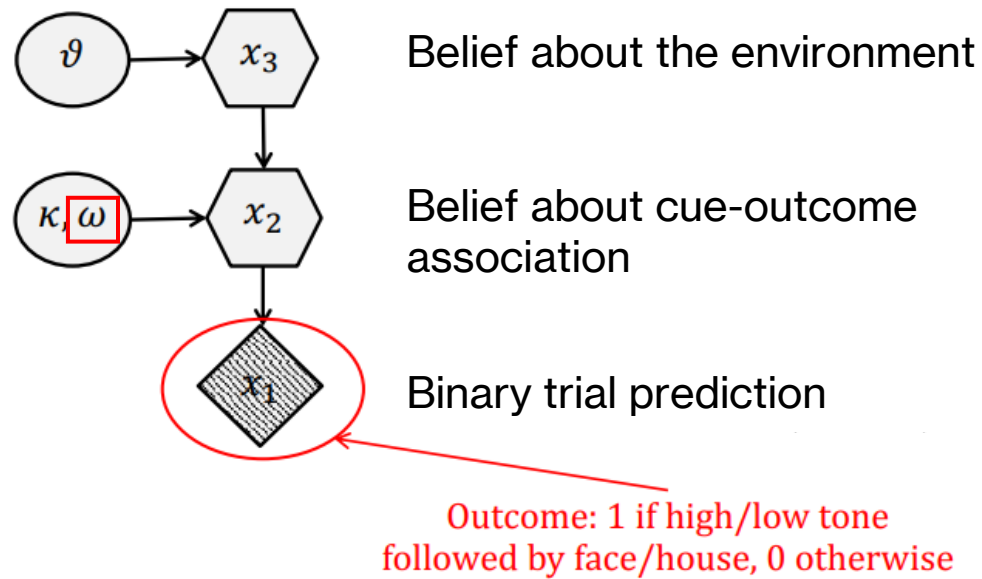
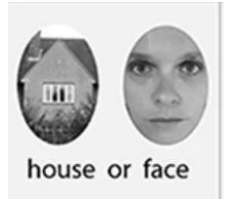
Relatively few errors overall

More errors when cue-outcome association are weak or unpredictable

More errors when visual noise is high but did not vary much

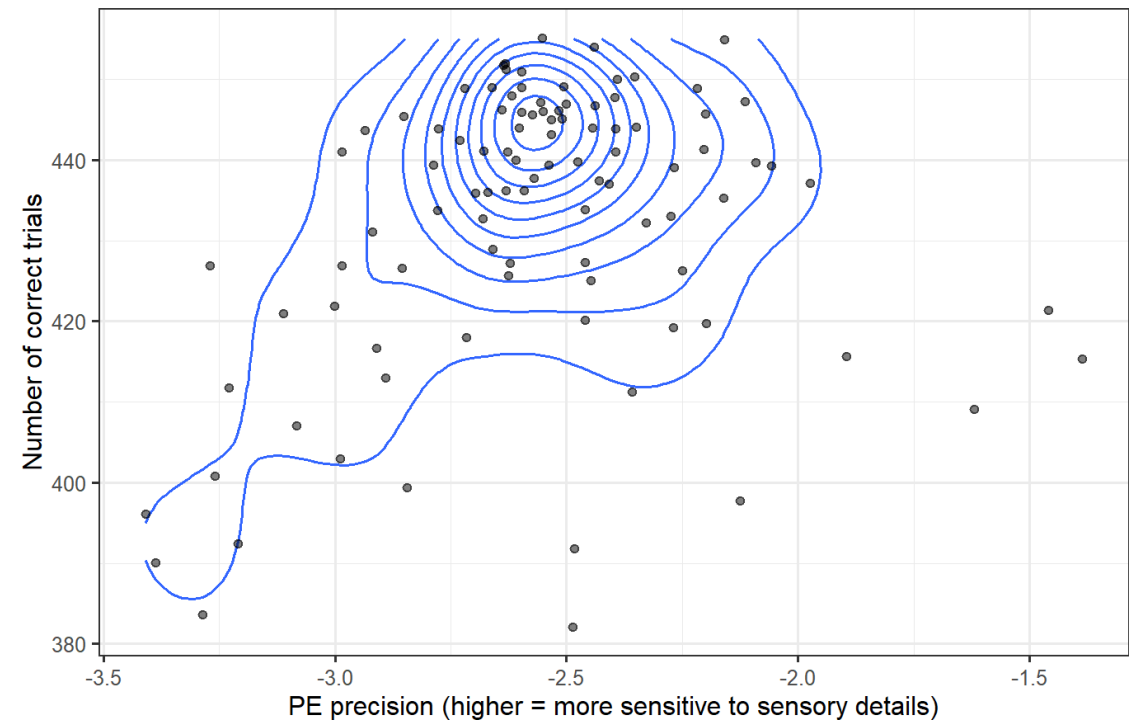
# Measuring sensitivity to sensory details

for each participant we fit a parameter reflecting this sensitivity

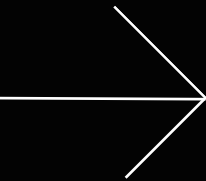


$\omega$  influences how much prediction error (PE) is weighted when  $x_2$  is updated at each time step; this is similar to a learning rate

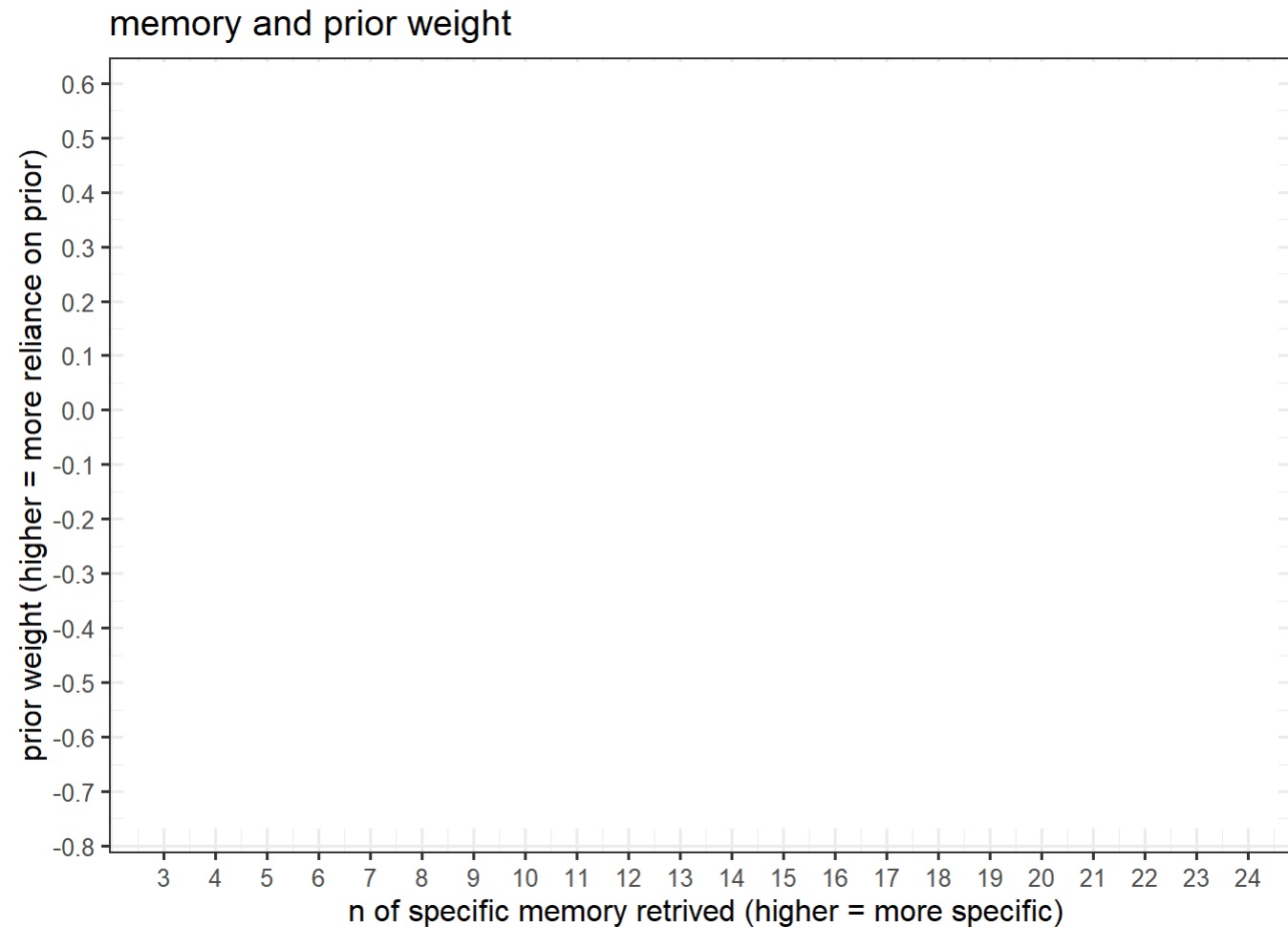
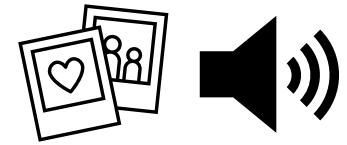
Density plot of PE precision and overall accuracy  
PE precision positively correlated with accuracy



SO, DO THEY CORRELATE?

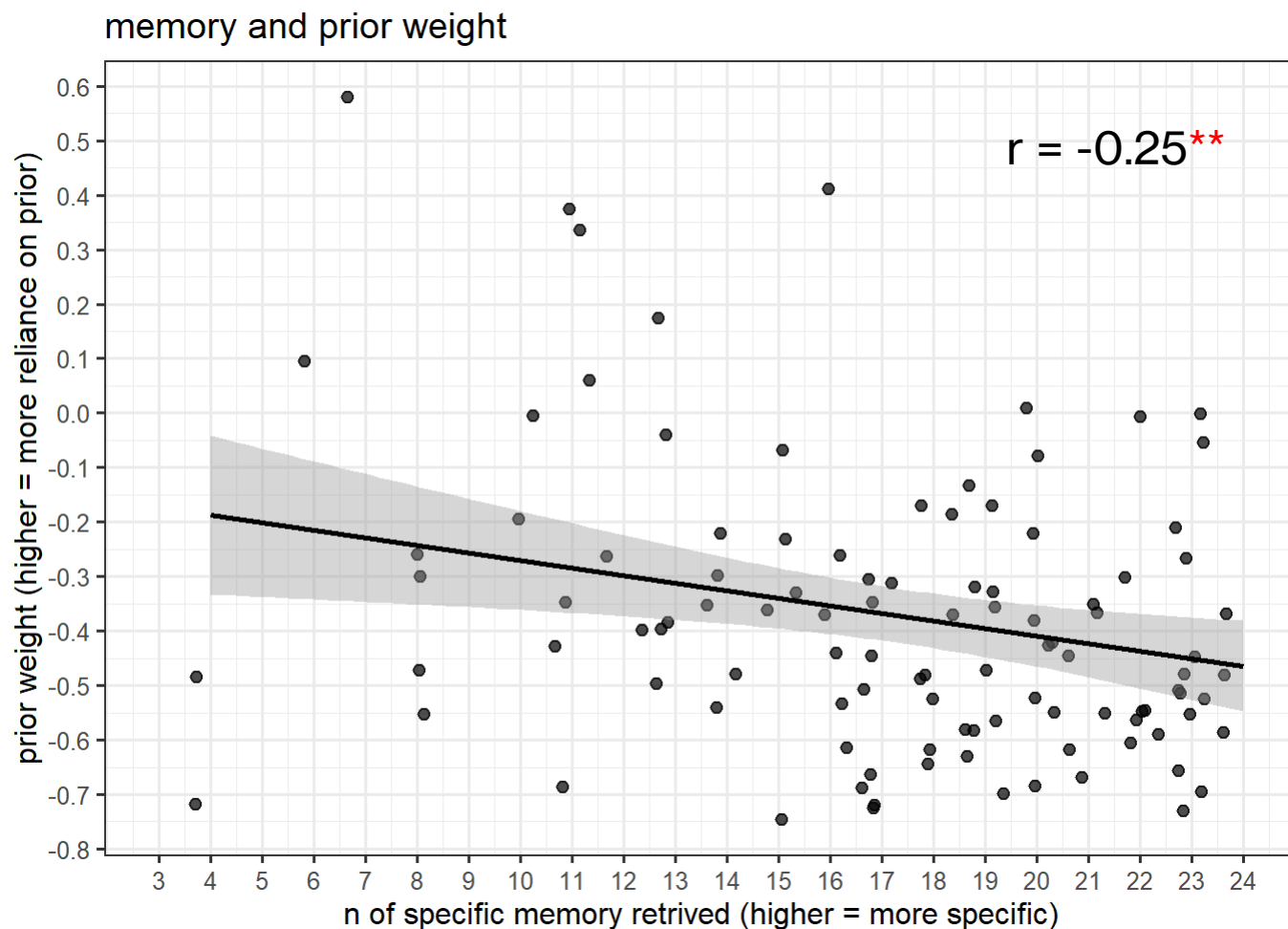
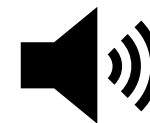
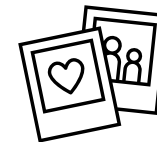


# SENTENCE PERCEPTION AND MEMORY SPECIFICITY



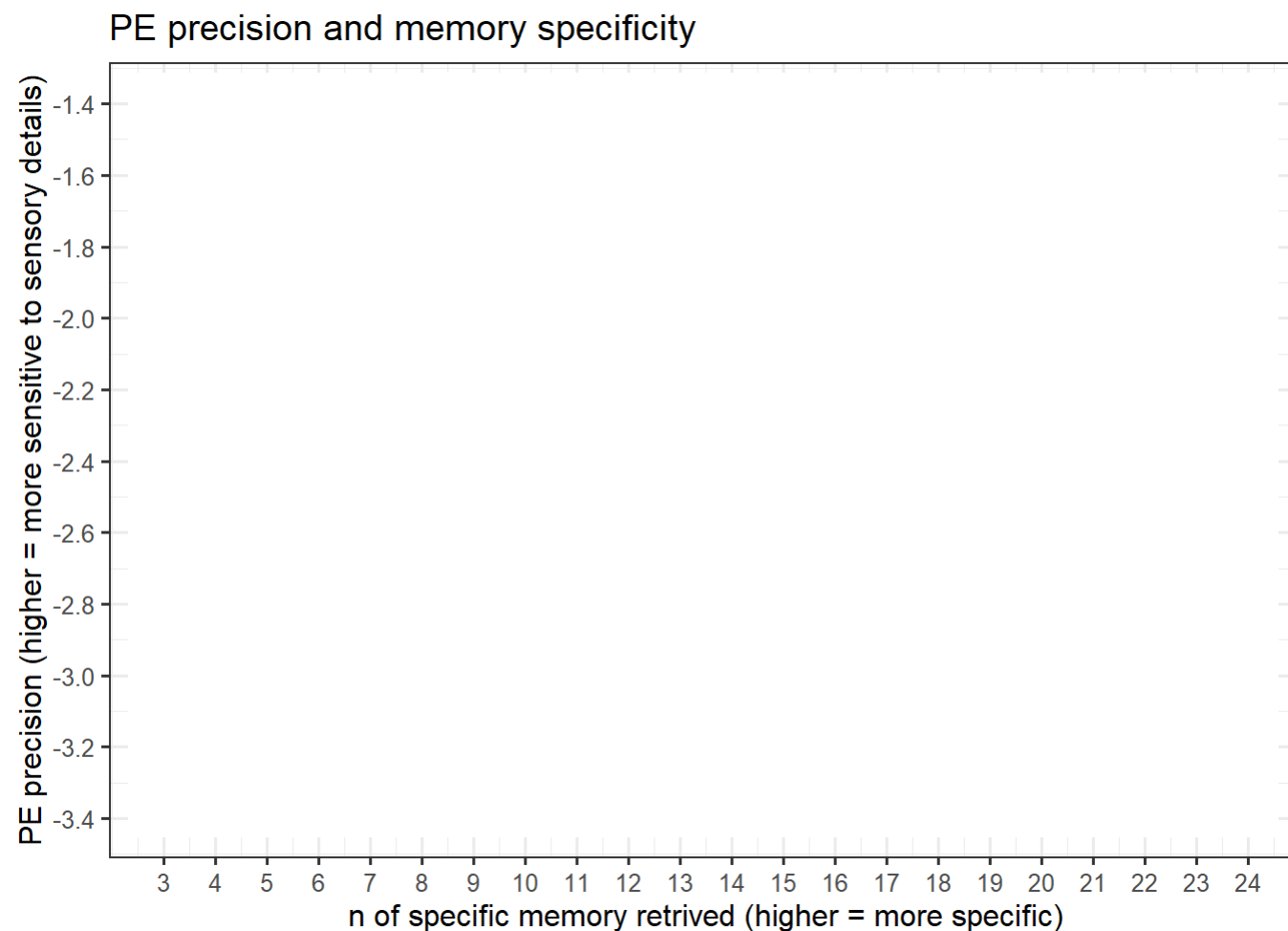
# SENTENCE PERCEPTION AND MEMORY SPECIFICITY

Memory specificity is negatively correlated with reliance on prior in sentence perception task



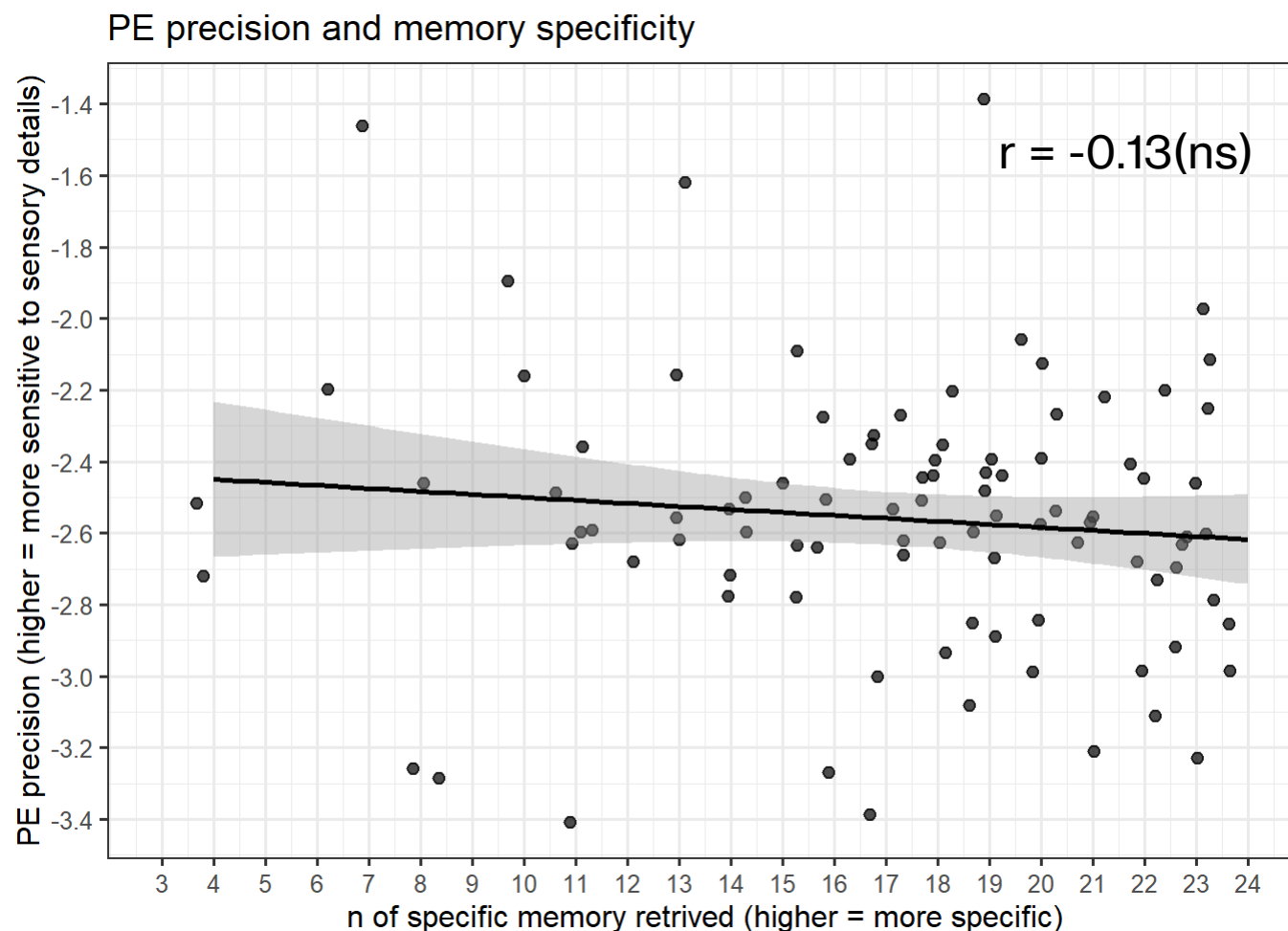
People with more specific memories rely on their prior knowledge less overall and perceive words more accurately during our task.

# PROBABILISTIC LEARNING AND MEMORY SPECIFICITY



# PROBABILISTIC LEARNING AND MEMORY SPECIFICITY

Memory specificity is not correlated with error precision in probabilistic learning task



Regardless of memory specificity, people seemed to learn new probabilistic associations equally well.

# Returning to our questions

**What if performance on the Autobiographical Memory Test partly captures a broader trade-off in the domain of memory?**

We have some evidence that people who rely more on sensory information during sentence perception retrieve more specific autobiographical events.

Some of the variation in memory specificity may be explained by these individual differences (though perhaps not the learning captured in the face/house task).

**Is this reliance on prior domain-general?**

Potentially not, but more work is needed

Maybe the reliance is different when prior knowledge is learnt versus new?

# Acknowledgement



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